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Additional SLH-DSA Parameter Sets for Limited-Signature Use Cases

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Initial Public Draft

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**Additional SLH-DSA Parameter Sets
for Limited-Signature Use Cases**

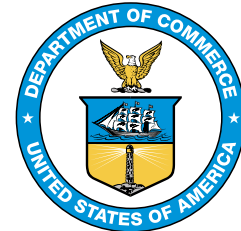
Initial Public Draft

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Abstract

This Special Publication serves as a technical extension of FIPS 205, the specification for the Stateless Hash-based Digital Signature Algorithm (SLH-DSA). While standard SLH-DSA parameter sets are designed to maintain full security for up to 2^{64} signatures, the resulting signature sizes and/or verification times are undesirable or prohibitive for certain applications. To address these performance constraints, this document specifies six additional SLH-DSA parameter sets for security levels 1, 3, and 5. These new variants are specifically tailored for use cases that require fast verification and reduced signature sizes, such as the signing of software, firmware, and digital certificates. These optimizations are achieved by establishing a strict limit of 2^{24} signatures per signing key. Because of this operational limit, these parameter sets are not approved for general-purpose use, and users must perform a thorough evaluation to ensure that the signature limit is never exceeded during any signing key's lifetime.

Keywords

cryptography; digital signatures; FIPS 205; hash-based signatures; post-quantum; post-quantum digital signatures; public-key cryptography; SLH-DSA; SPHINCS⁺; stateless hash-based signatures.

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The authors thank the cryptographic community for their insights regarding the design and security analysis of these limited-signature SLH-DSA parameter sets. The authors also thank the other NIST PQC team members for their input and specially John Kelsey for his insights.

Note to the Reviewers

NIST requests feedback on the suitability of these six additional SLH-DSA parameter sets. These variants are specifically tailored for applications that require fast verification and reduced signature sizes, such as signing software, firmware, and digital certificates. While

145 these optimizations are achieved by establishing a strict limit of 2^{24} signatures per signing
146 key, they offer significant performance advantages for those specific use cases.

147 Reviewers are encouraged to evaluate whether the operational requirements (e.g., assessing
148 the number of signing devices and the frequency of signature generation) are sufficient to
149 prevent exceeding the limit during any signing key's lifetime.

1. Introduction

Federal Information Processing Standards Publication (FIPS) 205 [1] specifies the Stateless Hash-based Digital Signature Algorithm (SLH-DSA), which is a mechanism derived from the SPHINCS⁺ submission to the NIST Post-Quantum Cryptography (PQC) Standardization Project. SLH-DSA is designed to provide digital signatures that are secure against potential future attacks by quantum computers, which would render many current public-key algorithms obsolete.

One of the primary advantages of SLH-DSA is that it is “stateless,” meaning it does not require the signer to maintain a record of the number of previously generated signatures to ensure security. To achieve this while maintaining full security for up to 2^{64} signatures per private key, the algorithm utilizes a large number of few-time signature keys. However, this high signature capacity results in significant performance trade-offs; for instance, the smallest signature sizes for security levels 1, 3, and 5 in the standard are 7,856, 16,224, and 29,792 bytes, respectively.

For many use cases of transmitting and verifying the digital signatures of firmware, software, and digital certificates, those signature sizes and/or their associated verification times are highly undesirable. This Special Publication (SP) addresses these constraints by specifying six new variants of SLH-DSA that are optimized for fast verification and reduced signature sizes. These improvements are made possible by establishing a lower limit of 2^{24} signatures per signing key.

The plan to standardize these six parameter sets is based on extensive research and community feedback [2–5]. Because of the 2^{24} signature limit, these parameter sets are specifically tailored for signing software, firmware, and certificates. To ensure the continued security of these variants, users **must** perform a thorough evaluation of their specific use case to ensure that the limit is never exceeded while considering additional factors, such as the number of devices sharing a signing key, signature generation frequency, and the intended operational lifespan of the key.

NIST may specify more parameter sets in the future if new needs arise.

2. Six Additional SLH-DSA Parameter Sets

This section introduces and approves six additional SLH-DSA parameter sets categorized across three distinct security categories: 1, 3, and 5. These parameter sets are specifically

designed for environments in which a signature limit of 2^{24} per signing key is acceptable. To achieve the desired performance optimizations for these use cases, this publication expands upon the parameter definitions found in Section 11 of FIPS 205. Specifically, the Winternitz parameter w is adjusted such that $w = 4$ ($lg_w = 2$) for security levels 1 and 5, and $w = 8$ ($lg_w = 3$) for security level 3, which differs from $w = 16$ ($lg_w = 4$) utilized in FIPS 205. These modifications allow for a strategic balance between signature size and verification speed.

2.1. SLH-DSA Parameter Sets at 2^{24} Signatures Limit Per Signing Key

The following six SLH-DSA parameter sets listed in Table 1 provide a specialized alternative to the parameter sets found in Section 11 of FIPS 205. These optimized variants achieve their performance gains by restricting the total number of signatures generated by a single private key to a maximum of 2^{24} signatures.

The trade-off for these performance benefits is a relatively slower signing process compared to the FIPS 205 parameter sets. Consequently, these parameters are best suited for "sign-once, verify-many" and other scenarios in which the computational cost of signing is less critical than the efficiency of the verification on end-user devices. Each parameter set is categorized into security levels 1, 3, or 5 to match the security strength requirements of different applications. Implementers should note that these specific configurations utilize adjusted Winternitz values, specifically $w = 4$ or $w = 8$.

	n	h	d	h'	a	k	lg_w	m	security category	pk bytes	sig bytes
SLH-DSA-SHA2-128-24 SLH-DSA-SHAKE-128-24	16	22	1	22	24	6	2	21	1	32	3,856
SLH-DSA-SHA2-192-24 SLH-DSA-SHAKE-192-24	24	21	1	21	25	9	3	32	3	48	7,752
SLH-DSA-SHA2-256-24 SLH-DSA-SHAKE-256-24	32	21	1	21	25	12	2	41	5	64	14,944

Table 1: SLH-DSA parameter sets at 2^{24} signature limit per signing key. The parameters $n, h, d, h', a, k, lg_w, m$ are defined in Section 11 of FIPS 205.

3. Usage and Security Requirements

The six parameter sets specified in this publication are specifically designed for applications that require high-performance verification and/or reduced signature sizes, such as signing software, firmware, and digital certificates. These optimizations are achieved by establishing a restricted signature limit. To maintain the intended security strength, it is a strict requirement that no more than 2^{24} signatures are generated using a single signing key. Because the security of these parameters depends entirely on adhering to this limit, users **must** perform a rigorous and complete evaluation of the operational environment for each signing key. This assessment **should** account for several critical factors, including the number of physical or virtual devices that share the same signing key, the maximum potential speed of each signing operation, the projected frequency of signature generation across all devices, and the total anticipated operational lifespan of the key. If an application's requirements or operational controls cannot guarantee that the 2^{24} signature limit will be respected, the parameter sets specified in this document **must not** be used.

To improve signing performance, an implementation may optionally cache some specific levels (a part) of the SLH-DSA hypertree or the whole hypertree. The storage of these cached values is permissible and does not affect the security properties of the algorithm.

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219 **Appendix A Acronyms**

220 SLH-DSA: stateless hash-based digital signature algorithm

221 FIPS: Federal Information Processing Standards

222 SP: Special Publication

223 SPHINCS⁺: Stateless Practical Hash-based Incredibly Nice Cryptographic Signature Plus

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